

|                    |                 |                        |                         |                                 |                         |                          |                           |                      |                     |                   |                                          |                                    |
|--------------------|-----------------|------------------------|-------------------------|---------------------------------|-------------------------|--------------------------|---------------------------|----------------------|---------------------|-------------------|------------------------------------------|------------------------------------|
| Material Selection |                 |                        |                         | Material Properties             | Material Properties     | Material Properties      | Material Properties       | Material Properties  |                     |                   |                                          |                                    |
|                    | Internal Layout |                        |                         |                                 | Component Inertia       |                          |                           | Component Inertia    |                     | Component Inertia |                                          |                                    |
|                    |                 | Planform Envelope      | Geometric Constraints   |                                 |                         | Planform Geometries      |                           | Planform Geometries  | Planform Geometries |                   | Planform Geometries                      |                                    |
|                    |                 |                        | Standard Planform       |                                 |                         | Planform Geometries      | Planform Geometries       | Planform Geometries  | Planform Geometries |                   | Planform Geometries                      | Planform Geometries                |
|                    |                 |                        |                         | LG Sizing                       | LG Inertia, Load Factor |                          |                           | LG Inertia           | LG Geometry         | LG Inertia        | LG Geometry                              |                                    |
|                    |                 |                        |                         |                                 | Fuselage Sizing         |                          |                           | Fuselage Mass        | Fuselage Geometry   | Fuselage Mass     | Fuselage Mass                            |                                    |
|                    |                 |                        |                         |                                 |                         | Wing Structural Analysis | Skin thickness            | Wing Mass            |                     | Wing Mass         | Wing Mass                                |                                    |
|                    |                 |                        | Skin Thickness, Taper   |                                 |                         | Skin Thickness           | Wing Aeroelastic Analysis |                      |                     |                   |                                          |                                    |
|                    |                 |                        |                         |                                 |                         |                          |                           | Empennage Sizing     | Empennage Geometry  | Empennage Mass    | Static Stability                         | Empennage Mass, Empennage Geometry |
|                    |                 |                        |                         |                                 |                         |                          |                           |                      | Drag Estimation     |                   | Thrust Required, Aerodynamic Performance | Thrust Required                    |
|                    |                 |                        |                         |                                 |                         |                          |                           |                      |                     | CG Movement       | Static Stability                         |                                    |
|                    |                 | Any Requirment Failed? | Any Requirement Failed? | Geometric Constraint Violation? |                         |                          |                           | Oversized Empennage? |                     |                   | Requirement Checks                       | Maximum Noise Exceeded?            |
|                    |                 |                        |                         |                                 |                         |                          |                           |                      |                     |                   | Maximum Noise                            | Noise and Emissions Estimation     |